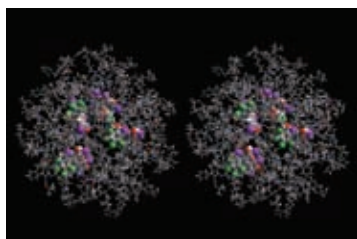


After much discussion and modeling, researchers determined that 60 carbon atoms form a single stable molecule only if they're arranged in 20 hexagons and 12 pentagons that are linked to form a sphere — as it happens, it's the same arrangement of hexagons and pentagons proposed by American architect and engineer Buckminster Fuller for his famed geodesic dome. In his honor these molecules got dubbed as buckminsterfullerenes.

Several applications are envisaged for fullerenes, such as miniature 'ball bearings' to lubricate surfaces, drug delivery vehicles and in electronic circuits. In accordance with this vision, several companies and universities have started developing methods for mass-manufacture of fullerenes. MIT, working with a company called Nano-C, Inc., developed a method called combustion synthesis, which produces big enough quantities of buckyballs — at a low enough cost — for use in commercial applications.

Dendrimers

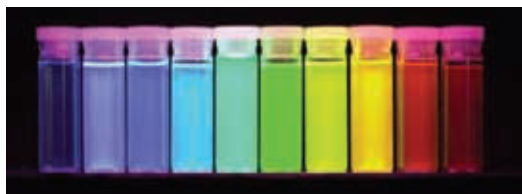
Dendrimers, formed through a nanoscale hierarchical self-assembly process, are spherical polymeric molecules. There are of many types - the smallest being several nanometres in size. Dendrimers are used in conventional applications such as coatings and inks, but they also have a range of interesting properties which could lead to useful applications. Environmental clean-up could be assisted by dendrimers as they can trap metal ions, which could then be filtered out of water with ultra-filtration techniques.



An intricate structure of Dendrimer

Quantum Dots

Quantum dots were theorized in the 1970s and initially created in the early 1980s. As we learned earlier, if these quantum dots (semiconductor particles)



Different colours exhibited by quantum dots

are made small enough, quantum effects come into play, which limit the energies at which electrons and holes can exist in the particles. As